Expt 3 Performance Analysis of IIR Filters

3.1 Design and Comparative Analysis of Butterworth and Chebyshev Type-I IIR Band-Pass Filters Using MATLAB

Code;'

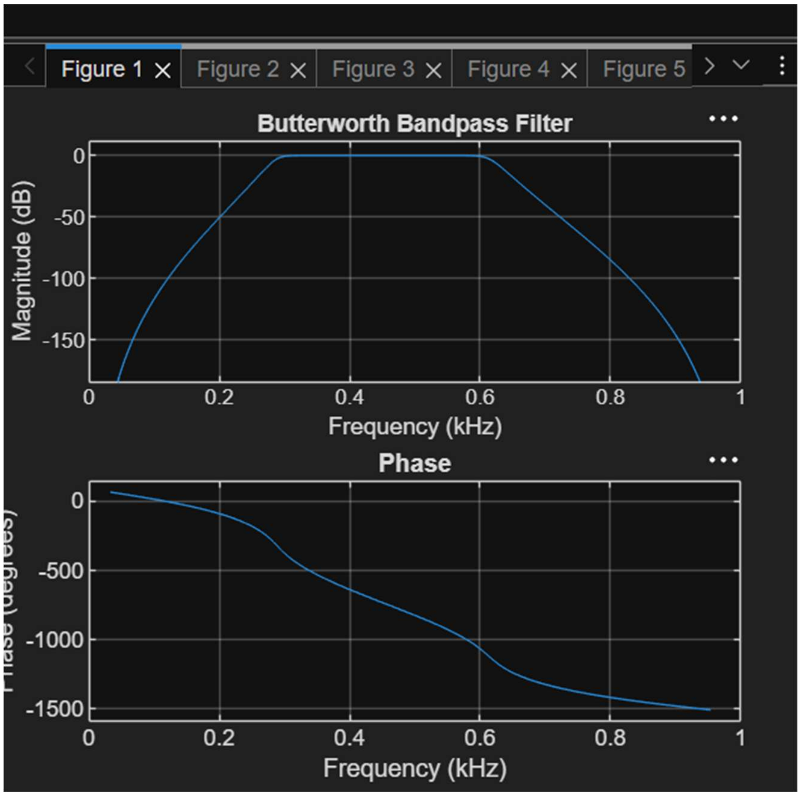
```
clc;
clear;
close all;

%% Filter Specifications
Fs = 2000;        % Sampling Frequency (Hz)
Rp = 1;           % Passband Ripple (dB)
Rs = 40;          % Stopband Attenuation (dB)

Wp = [300 600] / (Fs/2);   % Passband Edge Frequencies (Normalized)
Ws = [200 700] / (Fs/2);   % Stopband Edge Frequencies (Normalized)

%% Butterworth Bandpass Filter
[n1, Wn1] = buttord(Wp, Ws, Rp, Rs);
[b1, a1] = butter(n1, Wn1, 'bandpass');

%% Chebyshev Type-I Bandpass Filter
[n2, Wn2] = cheb1ord(Wp, Ws, Rp, Rs);
[b2, a2] = cheby1(n2, Rp, Wn2, 'bandpass');

%% Magnitude Response
figure;
freqz(b1, a1, 1024, Fs);
title('Butterworth Bandpass Filter');
```

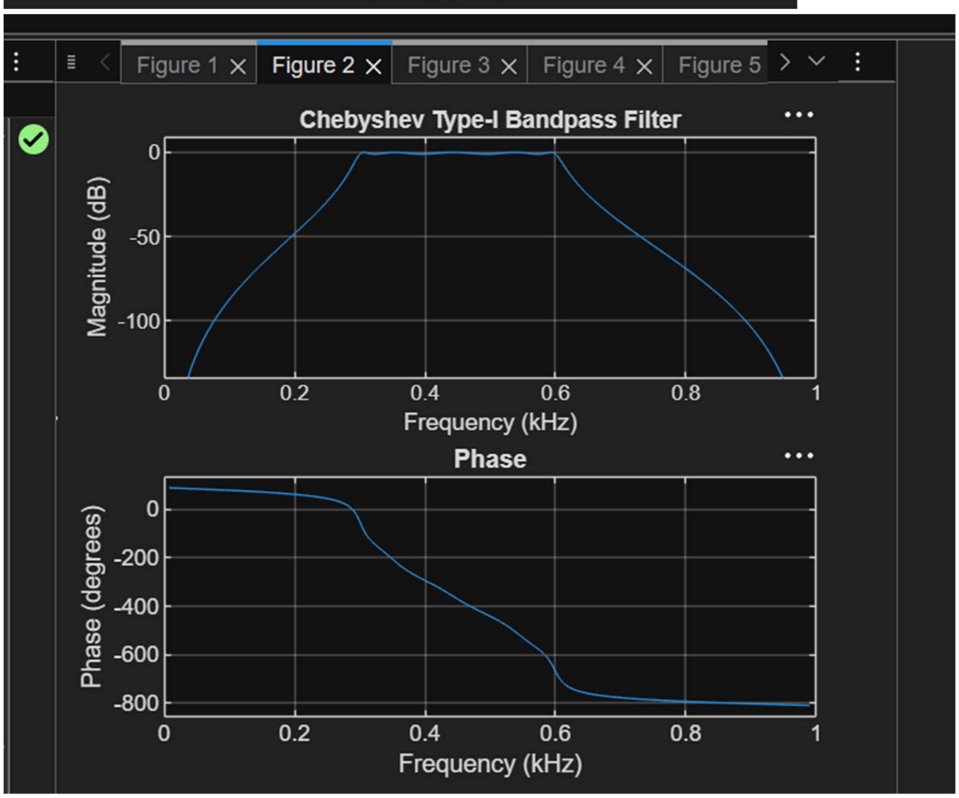
```
figure;
freqz(b2, a2, 1024, Fs);
title('Chebyshev Type-I Bandpass Filter');

%% Group Delay Comparison
figure;
grpdelay(b1, a1, 1024, Fs);
hold on;
grpdelay(b2, a2, 1024, Fs);
grid on;
legend('Butterworth', 'Chebyshev-I');
title('Group Delay Comparison');

%% Pole-Zero Plot
figure;

subplot(1,2,1);
zplane(b1, a1);
title('Butterworth Pole-Zero Plot');

subplot(1,2,2);
zplane(b2, a2);
title('Chebyshev-I Pole-Zero Plot');

%% Impulse Response
figure;
```

```
subplot(2,1,1);
impz(b1, a1);
grid on;
title('Butterworth Impulse Response');

subplot(2,1,2);
impz(b2, a2);
grid on;
title('Chebyshev-I Impulse Response');

%% Pole Magnitude
figure;

subplot(1,2,1);
stem(abs(roots(a1)), 'filled');
grid on;
xlabel('Pole Number');
ylabel('|Pole|');
title('Butterworth Pole Magnitude');

subplot(1,2,2);
stem(abs(roots(a2)), 'filled');
grid on;
xlabel('Pole Number');
ylabel('|Pole|');
title('Chebyshev-I Pole Magnitude');

%% Display Filter Orders
```

fprintf('Butterworth Filter Order = %d\n', n1);

fprintf('Chebyshev Type-I Filter Order = %d\n', n2);





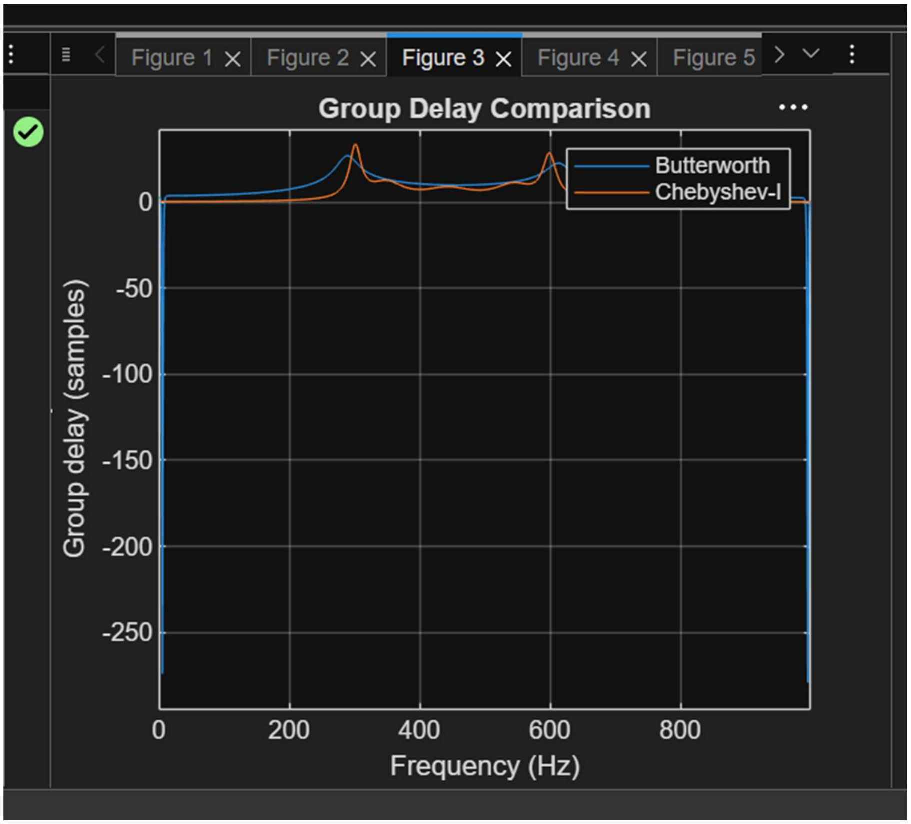


Figure 1
Figure 2
Figure 3
Figure 4
Figure 5
Group Delay Comparison
Butterworth
Chebyshev-I
0
-50
-100
-150
-200
-250
Group delay (samples)
0
200
400
600
800
Frequency (Hz)

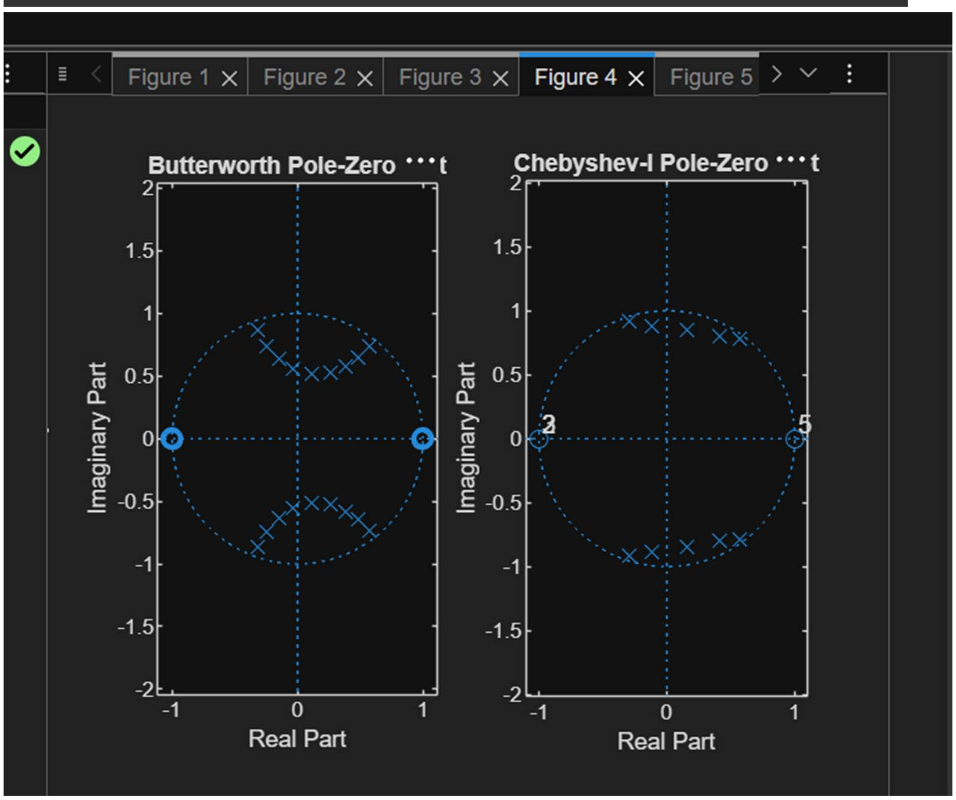
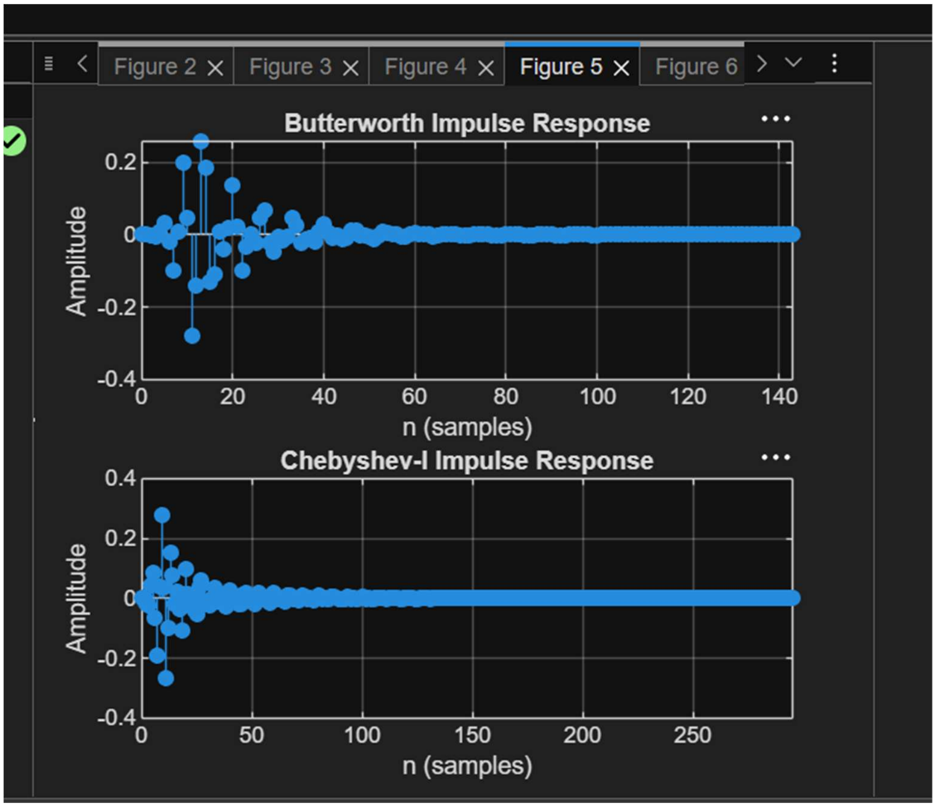
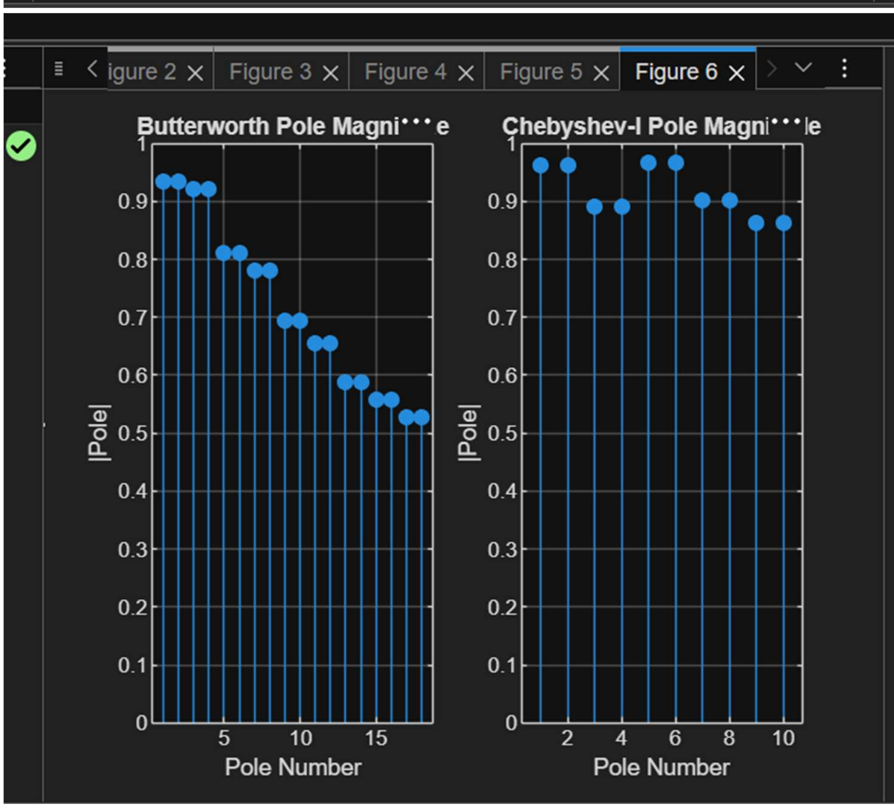


Figure 1
Figure 2
Figure 3
Figure 4
Figure 5
Butterworth Pole-Zero
Chebyshev-I Pole-Zero
Imaginary Part
Real Part
Imaginary Part
Real Part





Results: 1. Butterworth and Chebyshev Type-I IIR Band-Pass Filters were successfully designed and implemented in MATLAB. 2. The minimum filter orders and cutoff frequencies were obtained for both filters. 3. The frequency responses were analyzed and compared.